MILITARY SPECIFICATION

PROPELLANT M1
FOR USE IN CHARGE, PROPELLING, XM164

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M1 salted Propellant (see 3.1) for use in the XM164 Propelling charge employed with the 155MM Howitzers, M109A1 and XM198, during DT/OT II testing only.

1.2 Classification. The propellant shall be of the following types as specified:

Type I - Cylindrical Multi-perforated grains
Type II - Cylindrical Single-perforated grains

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-193 - Potassium Sulfate

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS

ARMY

9285574 - Charge, Propelling, XM164 (Green Bag) for 155MM Howitzer, XM198, M109A1

FSC: 1375

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington,D.C., 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M1 with 1.0 percent plus or minus 0.3 percent potassium sulfate (MIL-P-193) added, Type I for Zones 2 through 5 and Type II for Zone 1 conforming to the requirements of MIL-STD-652.

3.2 Proving Ground. The propellant,when fired with the weight of projectile (M549 (96.0 lb)) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.4.

3.2.1 Assessment. At 70 degrees Fahrenheit (°F) and standard weapon conditions, (with XM199 Cannon with at least 95% remaining life) the propellant(s) shall be capable of charge assessment to the velocities and pressures specified in Table I.

3.2.2 Uniformity of velocity. The standard deviation in velocity for the lot at Zone 2 and Zone 5 at 70°F shall not exceed 4.5 fps and 5.0 fps, respectively.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 26,000 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon (see 3.2.1).

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3.3 Workmanship. The best commerical practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

TABLE I

Velocity and Pressure

Zone	Proj	Velocity (fps)	Pressure(psi) Minimum(min)	Maximum(max)
1	M549	700	5200	6200
2	M549	850	-	-
3	M549	920	-	-
4	M549	1210	-	-
5	M549	1510	16000	22000

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality Conformance Inspection

4.3.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.3.2 Examination and Testing

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4.3.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.3.2.2 Proving ground (see 3.2). Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652.

4.3.2.2.1 Assessment (see 3.2.1). The lot shall be rejected if the assessed charge weight of propellant has an expected average pressure which does not meet the applicable requirements specified in Table I.

4.3.2.2.2 Velocity Uniformity (70°F) (See 3.2.2). If the product of the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.4.3 exceeds the applicable requirement the lot shall be rejected. For initial production lots a ten round sample shall be tested. (see 6.4)

4.3.2.2.3 Individual pressure (see 3.2.3)

4.3.2.2.3.1 Initial Production. If any individual pressure observed during uniformity firings at any operating temperature after correction for difference between as fired and recommended service charge weight exceeds 26,000 psi, the lot shall be rejected.

4.3.2.2.3.2 Subsequent Production 70°F. If any individual pressure observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 26,000 psi, the lot shall be rejected.

4.4 Test methods and procedures

4.4.1 Proving ground tests. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.4.2 Charge assessment

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4.4.2.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.1). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.4.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.4.2.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant-slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.4.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.4.3 Uniformity series

4.4.3.1 Initial Production

4.4.3.1.1 145°F. Ten rounds of XM164 Charges (Zone 5) shall be assembled with the assessed charge of propellant, conditioned to 145°F (as indicated by a thermocoupled control round (rnd)) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight). (see 6.7) (see 6.9)

4.4.3.1.2 70°F. Ten rounds of XM164 Charges each at Zone 1, Zone 2 and Zone 5 shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled

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control rnd) and fired alternately with calibration rounds. The velocity and pressure means, as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to standard shell weight and standard weapon/ firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions) and to recommended charge weight conditions. (see 6.9)

4.4.3.1.3 -50°F . Ten rounds of XM164 Charges (Zone 5) shall be assembled with the assessed charge of propellant, conditioned to -50°F (as indicated by a thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight). (see 6.9)

4.4.3.2 Subsequent Production. Seven rounds each of Zone 1, Zone 2 and Zone 5 Cartridges shall be assembled and tested at 70°F while 145°F and -50°F testing shall be omitted entirely. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results at 70°F shall be corrected to standard shell weight (M549 (96.0 lb)) and standard weapon/firing conditions and recommended charge weight conditions (see 4.3.2.2.3.2)

4.4.4 Refiring. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering Data. Ordering data shall be in accordance with MIL-A-48078 (PA) with the following exceptions.

6.1.1 Procurement Requirements. Procurement requirements of MIL-A-48078(PA) shall apply except for First Article Sample.

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6.1.2 Contract Data Requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-AD-E-P-2, Dover, New Jersey 07801.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more or failure of a lot shall cause testing to revert back to initial production plan.

6.5 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.6 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SARPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.7 Test Charges. Test charges shall be assembled in accordance with drawing 9285574.

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6.8 Proving ground test summary

Initial Production

Temp	Sample Size	Vel(fps) (assessed)	Lot Std. Dev of Vel (fps)	Perm Ind Max Pres (psi)	Avg Pres. Max (psi)
-50	10(Zone 5)	n/a	n/a	26,000	n/a
70	10(Zone 1)	700(M549)	n/A	26,000	6,200
70	10(Zone 2)	850(M549)	4.5	26,000	-
70	10(Zone 5)	1510(M549)	5.0	26,000	22,000
145	10(Zone 5)	n/a	n/a	26,000	n/a

Subsequent Production

70	7(Zone 1)	700(M549)	n/a	26,000	6,200
70	7(Zone 2)	850(M549)	4.5	26,000	-
70	7(Zone 5)	1510(M549)	5.0	26,000	22,000

6.9 Calibration rounds. When calibration rounds are not available for DT/OT II tests, the first propellant lot assessed shall be utilized as the reference propellant for subsequent propellant lot assessments. All assessments performed without calibration rounds shall be conducted with a 25 round sample at each temperature in lieu of the specified 10 round sample. Every effort shall be made to avoid component substitution when the first propellant lot is used as reference. Assessments conducted without calibration rounds shall be referred to Picatinny Arsenal, ATTN: SARPA-QA-A-P for disposition.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-089

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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